Step 10: Single-Stage Preference Alignment (ORPO) and Verification

10.1 Single-stage reference-free preference alignment formulation (ORPO):

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda_{\text{OR}} \cdot \mathcal{L}_{\text{OR}} \quad \text{where} \quad \mathcal{L}_{\text{OR}} = -\log \sigma \left(\log \left[\frac{\text{odds}_\theta(y_w \mid x)}{\text{odds}_\theta(y_l \mid x)} \right] \right)$$

Where: $\mathcal{L}_{\text{ORPO}}$: Total loss | $\mathcal{L}_{\text{SFT}}$: SFT cross-entropy loss | λ_{OR}: Odds weight | $\mathcal{L}_{\text{OR}}$: Odds ratio penalty | $\text{odds}_\theta(y \mid x) = \frac{P_\theta(y|x)}{1-P_\theta(y|x)}$: Output sequence odds | y_w, y_l: Chosen / rejected responses for prompt x.

10.1.1 Jointly maximize chosen likelihood while penalizing rejected tokens (2× memory savings).

10.1.2 Supervise via UltraFeedback pairs $\mathcal{D} = \{(x, y_w, y_l)\}$, bypassing unstable RLHF loops.

10.2 Enforce runtime quality gating via preference score barrier $\mathcal{S}_{\text{ORPO}}(x, y) \geq \tau_{\text{approval}}$:

10.2.1 Audit deliverables using fine-tuned model (`reward_orpo_v2`, Qwen2.5-1.5B LoRA).

10.2.2 Score candidate outputs on instruction adherence, structural constraints, and safety guardrails.

10.3 Handle quality failures via targeted DAG re-planning:

10.3.1 Capture diagnostic error traces and schema deviations when output falls below τ_{approval}.

10.3.2 Dispatch error logs to DAG Planner for targeted branch re-planning without system abort.